Renal, Nephrology & Transplant

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 88	Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter
Ch 90	Fundamentals of Nephrology
Ch 91	Renal Transplantation
★	Putting it together: kidney function and transplant decisions

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

1 · Kidney: position and relations

The kidney's neighbours are what you injure at surgery.

- The kidney is ≈**11 cm** long, 125–170 g (male), at T12–L3. The **right lies 1–2 cm lower**; both descend ≈3 cm on inspiration or standing.
- P1 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal mass — *relations, coverings, Gerota open below, hilum order*

Not in the paper answer:

- Ligament traction tears the liver (hepatorenal) or splenic (splenorenal) capsule.
- Posteriorly also lie the iliohypogastric and ilioinguinal nerves and pleura.
- P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — *access hazards: pleura, lung, liver/spleen, retrorenal colon*

Not in the paper answer:

- Upper-pole excision through the diaphragm causes pneumothorax.

2 · Coverings and compartments

Fascial spaces decide where fluid spreads.

- Perirenal fat enters the sinus. Each perirenal space is separate, so extravasation rarely crosses.
- The **anterior pararenal space** (colon, duodenum, pancreas) crosses the midline; the posterior pararenal space holds fat only.
- Transperitoneally, open the **white line of Toldt**. Reach the right hilum by a **Kocher manoeuvre**; on the left, reflect pancreatic tail, spleen and splenic vessels medially.

3 · Parenchyma and its imaging

Knowing the normal appearance lets you spot disease.

- Cortex is ≈1 cm. **Columns of Bertin** carry interlobar arteries, so the safe track is **through pyramid and papilla**.
- Glomerular filtration rate (GFR) is ≈**125 mL/min**; ≈99% of filtrate is reabsorbed. Urothelium lines calyces to urethra, so urothelial carcinoma arises anywhere.

Modality	Normal kidney
Ultrasound	Cortex iso-/hypoechoic to liver (hyperechoic in newborn)
Computed tomography (CT)	Unenhanced 30–60 Hounsfield units (HU), cortex = medulla; enhanced 80–120 HU
Magnetic resonance (MRI)	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**

CT phase	Timing	Use
Excretory	>3 min	Collecting system, ureter

- Intravenous urography: nephrotomogram 60–90 s, panoramic 15 min; no excretion at 24 h means a non-functioning kidney.
- **Corticomedullary differentiation** is lost in chronic kidney disease and acute kidney injury and returns with recovery. Parenchymal volume tracks function better than length.

Congenital anomalies

- **Malrotation**: central pelvis; calyces medial to it are the hallmark.
- **Ectopia**: ureter length fits the position, but poor drainage causes stasis, infection and stones.
- **Crossed fused ectopia**: the crossed ureter returns to a normal orifice.
- **Horseshoe**: lower poles fuse and ascent is arrested by the inferior mesenteric artery (**IMA**); it brings ureteropelvic junction obstruction (UPJO), stones, infection and multiple arteries.

4 · Renal vessels, lymphatics and nerves

End arteries make partial nephrectomy possible.

- P2 Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment — *segmental end arteries, accessory 25–28%, Brödel line, veins; left renal vein **6–10 cm**, nutcracker, circumaortic*

Not in the paper answer:

- The Brödel line site varies, so map it by angiography or methylene blue (Boyce anatrophic nephrolithotomy) *see Ch 91*.
- **CT angiography** is the gold standard for vascular anatomy but misses vessels <2 mm.
- Left lymph goes to left para-aortic nodes; right to interaortocaval/paracaval, but **may cross to left para-aortic**.

5 · Percutaneous puncture and vessels

The puncture site decides the bleeding risk.

- P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — *Sampaio puncture table: upper infundibulum 67%, fornix no arterial injury*

Not in the paper answer:

- A through-and-through puncture hits anterior segmental vessels and causes marked haemorrhage.

> **KEY POINT**
> Why the fornix? Arteries run on the anterior and posterior faces of the infundibula, and a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Vascular injury causes haemorrhage, arteriovenous fistula, pseudoaneurysm and loss of function.

6 · Pelvicalyceal system

Calyceal pattern decides what one puncture can reach.

- P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — *Sampaio types, duplex pelvis, perpendicular and crossed calyces*

Not in the paper answer:

- An interpelviocalyceal region on fluoroscopy is the indirect sign of crossed calyces.

- Dilatation is mimicked by an extrarenal pelvis, **megacalycosis** (non-obstructive, extra calyces, normal pelvis) and calyceal diverticulum (fornix into cortex).

7 · Ureter: course and relations

Injuries occur at crossings; stones stick at narrowings.

- The adult ureter is 22–30 cm (neonate 6.5–7 cm). Stones impact at three **narrowings**: ureteropelvic junction (UPJ), iliac crossing and ureterovesical junction (UVJ).
- It lies on psoas and crosses the **common iliac bifurcation** (landmark). Gonadal vessels cross in front; ileum and appendix lie near the right, sigmoid mesocolon near the left. The **pelvic brim** is a commonest injury site.
- In females it bounds the ovarian fossa and passes under the **uterine artery 1.5–2 cm** from the cervix ("water under the bridge"); the left is more at risk in vaginal surgery. In males the vas loops over it.
- The **intramural** segment is 1.2–2.5 cm in Waldeyer sheath. Bladder pressure closes this oblique tunnel, which prevents reflux.

8 · Radiologic anatomy of the ureter

Several normal findings mimic obstruction.

- Because of **peristalsis**, the whole ureter is rarely opacified; a continuous column suggests obstruction or atony.
- A stricture needs **proximal dilatation**; crossing vessels and atherosclerotic iliac indentation mimic it. Prostatic enlargement makes a "fish-hook" distal ureter.
- Incomplete duplication (one bud splits, one orifice) is commoner than complete (two buds, two orifices).

> **KEY POINT**
> **Weigert–Meyer rule**: the upper-moiety orifice is **inferior and medial** and obstructs (ureterocele); the lower moiety refluxes (vesicoureteral reflux, VUR).

- Bilateral medial displacement: retroperitoneal fibrosis, pelvic lipomatosis, after abdominoperineal resection. Unilateral: hypogastric nodes or aneurysm, bladder diverticulum.

9 · Ureteric blood supply, nerves, histology

Artery entry side tells you where to cut safely.

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- Lymph: abdominal ureter to para-aortic/paracaval, pelvic ureter to iliac nodes.
- The longitudinal anastomosis is fragile, so preserve adventitia and avoid lateral retraction to prevent ischaemic stricture.
- Peristalsis is **myogenic** (minor-calyx pacemakers): an excised ureter still contracts, and denervation does not cause reflux. Pain (T11–L1; S2–S4 lower) refers to flank, groin, scrotum

or labia.

- In **UPJO**, the junction has excess longitudinal muscle and collagen.

10 · Endoscopic anatomy

Landmarks guide safe ureteroscopy.

- The trigone (base: interureteric ridge, Mercier bar) is the bladder's most vascular part. Orifices lie 1–2 cm from midline and move laterally on filling.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- A higher grade means more reflux. The **intramural ureter is narrowest** (3–4 mm) and may need dilatation.
- Brim and UPJ look stenotic but open with irrigation; iliac pulsation marks the brim, where the ureter angles back.
- The lower-pole infundibulum needs ≈**140°** deflection. Ducts of Bellini dilate with obstruction.

ALGORITHM

1. Infundibulotomy: incise superior, then inferior, then posterior quadrants.
2. **Never anterior**.

90 Fundamentals of Nephrology

1 · Why a urologist needs nephrology

Urological surgery changes kidney physiology.

- Each kidney has ≈**1 million** **nephrons**, fixed at birth. When nephron mass falls (nephrectomy, agenesis), the rest **hyperfilter**.
- Interstitial fibroblasts make **erythropoietin** in hypoxia; the proximal tubule makes **calcitriol** by 1α-hydroxylation, driven by parathyroid hormone (PTH) and low calcium or phosphate.

> **KEY POINT**
> Surgery, obstruction, its relief and drug dosing all change volume, electrolytes and acid–base. The glomerulus is a pressure filter set by arteriolar tone; the tubule is energy-hungry and near-hypoxic. Most disorders fail one or the other.

2 · Measuring kidney function

Serum creatinine (SCr) often misleads in surgical patients.

- P2 GFR estimation: clinical assessment and in the prospective transplant donor — *inulin/iohexol gold standard; creatinine secreted (24-h clearance overestimates); CKD-EPI 2021 race-free; cystatin C; combined equation most accurate*

Not in the paper answer:

- The old race coefficient raised estimated GFR (eGFR) in Black patients, delaying referral.
- SCr is falsely low with small muscle mass or amputation.
- Use the combined equation when creatinine eGFR is biased or a decision sits at a staging threshold.
- SCr is falsely raised by muscle, meat, creatine, and by **trimethoprim, cimetidine** and fenofibrate (block secretion). Cystatin C is raised by hyperthyroidism, steroids and inflammation.

> **KEY POINT**
> Trap: after cystectomy or bladder repair, a rising SCr with ascites may be an **intraperitoneal urine leak** reabsorbed across the peritoneum, not kidney injury.

- P1 Proteinuria — *normal <**150 mg/day**, nephrotic >3.5 g/day; spot early-morning ACR first, PCR for non-albumin protein; dipstick misses light chains*

Not in the paper answer:

- Albuminuria independently predicts chronic kidney disease (CKD) progression, cardiovascular disease, stroke and death.
- **eGFR is invalid when function is changing** (an anuric patient has GFR zero). So acute kidney injury (AKI) is defined by SCr change and urine output; both are late, and novel biomarkers are investigational.

3 · Acute kidney injury

Most surgical AKI is preventable or reversible.

Glomerular haemodynamics

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32.

Arteriolar change	Renal blood flow	GFR
Afferent constriction	Falls	Falls
Afferent dilatation	Rises	Rises
Efferent constriction	Falls	Rises
Efferent dilatation	Rises	Falls

Efferent constriction lowers flow but raises GFR, so angiotensin II blockade drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥ **0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- P2 Acute renal failure (AKI) and acute tubular necrosis — *causes, ATN mechanism, AEIOU dialysis indications*

Not in the paper answer:

- The GFR fall in ATN is protective, cutting tubular workload during regeneration.
- Muddy-brown casts are sloughed cells plus Tamm–Horsfall protein.
- **3 months** separates acute kidney disease (AKD) from CKD. Intratubular causes: myeloma casts, crystals.

Prerenal AKI

- Urine is bland, fractional excretion of sodium (FeNa) is low, and function **recovers immediately** on correction.
- **Autoregulation**: the **myogenic reflex** dilates the afferent arteriole via prostaglandins, so it is **disabled by NSAIDs** (non-steroidal anti-inflammatory drugs).
- In **tubuloglomerular feedback (TGF)**, low macula densa NaCl releases renin, and angiotensin II constricts the efferent arteriole, so it is **disabled by renin–angiotensin–aldosterone system (RAAS) blockade**.
- Age, CKD, transplant, sepsis and contrast impair autoregulation. **Modifiable** factors: hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis, hypercalcaemia.
- P2 Acute renal failure (AKI) and acute tubular necrosis — *prerenal vs ATN indices: urine Na 20, FeNa <**1%**, FeUrea <**35%**, SG, muddy-brown casts, fluid response*

Not in the paper answer:

- Heart failure needs diuretics; postoperative hypotension, holding antihypertensives.
- High FeNa after diuretics does not prove ATN; use fractional excretion of urea (FeUrea). **Response to correction is the single most important test.**
- Keep mean arterial pressure (MAP) > **65 mm Hg**; elderly vasculopaths need more.

Fluids (mOsm/L): plasma and 0.9% saline 308; Plasmalyte 294; Ringer's lactate 273; 0.45% saline 154; 5% dextrose (D5) 0, so D5 is free water.

Intrinsic AKI: the cast localises the compartment

	Acute tubular necrosis (ATN)	Acute glomerulonephritis (GN)	Acute interstitial nephritis (AIN)
Cause	Ischaemia, toxins	Immune complex, anti-glomerular basement membrane (anti-GBM), cell-mediated	Drug, infection, systemic
Urine	Granular casts	Dysmorphic red cells, **red cell casts**, proteinuria	White cells, **white cell casts**, with or without eosinophils

Acute tubular necrosis

- **Rhabdomyolysis** (obese patient, long flexed lateral robotic case; may be painless): give **early aggressive isotonic fluid**; alkalinisation is unproven. †Prevent it with padding and by limiting flexion and time.
- **Give diuretics only for symptomatic overload; they do not treat ATN.**

Postcontrast AKI

- This is an SCr rise within **48 h** of intravenous (IV) iodinated contrast. Risks are age, diabetes, CKD, anaemia and the arterial route.
- **IV fluid before contrast is the only effective prevention** (†for eGFR <30 or AKI); use the lowest dose. Post-contrast dialysis is never needed.
- **Gadolinium** groups I and III cause nephrogenic systemic fibrosis (**NSF**), so avoid them in AKI, eGFR < **30** and dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) is allowed. **Group III (gadoxetate) is treated as group I.**

Acute interstitial nephritis

- **Drugs cause most**: β-lactams, quinolones, sulfonamides, vancomycin, NSAIDs, proton pump inhibitors, allopurinol and **checkpoint inhibitors** (AIN is the commonest biopsy lesion in their AKI).
- Onset is days to weeks after the drug; suspect it in unexplained hospital AKI with **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ.
- **Stop the drug**; early steroids help (observational); ≈half recover fully.

Glomerulonephritis

- In microhaematuria, a rising SCr, proteinuria, red cell casts, hypertension or purpura call for **urgent nephrology and biopsy**, not cystoscopy first (rapidly progressive GN). Nephrotic disease is insidious.

Complications of AKI

- **Overload** is urgent only with respiratory distress. Restrict salt and fluid, give high-dose IV loop diuretic redosed by half-life, and add metolazone or acetazolamide (**sequential blockade**).
- Oral furosemide needs twice the IV dose; torsemide lasts longest (8–12 h).
- Uraemic bleeding: desmopressin (**DDAVP**) **0.3 µg/kg** IV/IM 30–60 min before the procedure; tachyphylaxis after 3 doses; watch sodium.

Renal replacement therapy

- AEIOU (acidosis, electrolytes, ingestions, overload, uraemia) means failed medical management, not a biochemical threshold. **Diffusion** clears solute; **convection** removes fluid. **Early dialysis gives no benefit, and continuous therapy is not superior.**

- Non-recovery for ≥**90 days** means CKD or end-stage kidney disease (ESKD).

4 · Drug dosing in reduced kidney function

Wrong dosing in renal impairment causes toxicity.

- Dose on steady-state eGFR; in AKI it overestimates, so reassess daily.
- †**Cockcroft–Gault** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers in dehydration or perioperative hypotension ("sick-day" rule).
- †Avoid nitrofurantoin if eGFR <30 (UK guidance allows short courses at 30–44). †Stop metformin at eGFR <30, and withhold it before iodinated contrast in AKI or eGFR <30.

5 · Chronic kidney disease

CKD alters surgical risk and planning.

- **KDIGO** (Kidney Disease: Improving Global Outcomes) defines CKD as abnormal structure or function for ≥**3 months**. Structure alone qualifies (atrophic kidney, normal GFR). Classify by **cause, GFR and albuminuria**.

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 count as CKD only with albuminuria or structural damage. Albuminuria can shift risk two bands without any GFR change: eGFR 43 with ACR 1100 mg/g is **G3bA3, very high**.

- Causes: diabetes, hypertension, GN, **stones, chronic obstruction**, genetic. **Each perioperative AKI is a modifiable progression factor**, as are albuminuria, smoking and acidosis.
- P2 Metabolic abnormalities in CKD — *complications table: anaemia (Hb **10–11.5 g/dL**, >13 harmful), secondary/tertiary HPT, acidosis (bicarbonate ≥**22**), hyperkalaemia (binders, keep RAAS blockade), hypertension (chlorthalidone), ED*

Not in the paper answer:

- Progression is non-linear: sudden drops follow sepsis, nephrotoxins and surgical complications.
- Calculators predict dialysis after nephrectomy from baseline function.
- Tertiary hyperparathyroidism (HPT): calcimimetic, then subtotal parathyroidectomy. Erectile dysfunction: phosphodiesterase-5 inhibitors, transplant.

Slowing progression

- Target glycated haemoglobin <7.5% and blood pressure <**130/80**; the agent matters only in proteinuric disease (RAAS blockade).
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors** protect kidney and heart and cut mortality, with or without diabetes. Add **finerenone** in CKD with type 2 diabetes.

6 · Haematuria and proteinuria: when to involve a nephrologist

Glomerular bleeding needs a nephrologist, not just cystoscopy.

- †Microhaematuria is ≥**3** red blood cells per high-power field (RBC/HPF) on microscopy; a positive dipstick needs confirmation.
- **Glomerular clues**: dysmorphic red cells, red cell casts, raised ACR or protein-to-creatinine ratio (PCR), rising SCr, hypertension, and purpura or systemic disease. †Glomerular disease does not exclude urothelial cancer.

ALGORITHM

1. Haematuria with or without proteinuria: send urine microscopy, ACR, SCr/eGFR.
2. If there are casts, dysmorphic cells, AKI or systemic features, then refer for **urgent nephrology**.
3. If isolated non-glomerular haematuria, then urological evaluation.
4. If isolated albuminuria, then stage CKD and refer per criteria.

- †KDIGO referral criteria: AKI or an abrupt GFR fall; GFR <30; ACR ≥300 mg/g (≥30 mg/mmol); a fall >5 mL/min/1.73 m²/year; red cell casts or haematuria >20 RBC/HPF; BP refractory to ≥4 drugs; persistent potassium abnormality; recurrent stones; hereditary disease.

7 · Electrolyte and acid–base disorders

Irrigation, diversions and obstruction upset salt, water and acid.

Water and sodium

- **Sodium disorders are water disorders**; volume status is a separate clinical judgement.
- Antidiuretic hormone (ADH) acts on V2 receptors to insert aquaporin-2 (**AQP2**). **Volume depletion overrides hypo-osmolality**.
- Of 1 L, D5 leaves **83 mL** in the circulation, 0.9% saline 250 mL and 5% albumin 1000 mL.

Hyponatraemia

- Confirm hypo-osmolality; correct sodium for hyperglycaemia. †Transurethral resection (TUR) syndrome is absorption of hypotonic glycine or sorbitol during monopolar resection.

Volume	Urine	Causes
Hypovolaemic	Na >20	Diuretics, mineralocorticoid deficiency, cerebral salt wasting
Hypovolaemic	Na <20	Vomiting, diarrhoea
Euvolaemic	Osm > serum	Syndrome of inappropriate ADH (SIADH), cortisol deficiency, hypothyroidism, pain, drugs
Euvolaemic	Osm low	Polydipsia, low solute ("tea and toast", beer potomania)
Hypervolaemic	Na >20	Kidney failure
Hypervolaemic	Na <20	Heart failure, cirrhosis, nephrotic

- SIADH is diagnosed by exclusion; treat with water restriction, then vaptans, salt with furosemide, or urea.
- **Rate**: in chronic hyponatraemia, raise sodium ≤**8–12 mEq/L per 24 h**, or **osmotic demyelination** follows.

- Risk is highest after resuscitated hypovolaemia (water diuresis); treat overcorrection with DDAVP and free water.
- With severe symptoms, give **3% saline to raise sodium 4–6 mEq/L**, then slow down.

Hypernatraemia

- Nephrogenic diabetes insipidus (DI) follows lithium, hypercalcaemia, hypokalaemia and **relief of obstruction**. Nil-by-mouth DI patients are at risk.
- Give free water. In chronic cases, lower sodium ≤ **0.5 mEq/L/h**, or cerebral oedema follows.

Potassium

- P1 Management of acute hyperkalaemia — *98% intracellular; causes incl. obstructive uropathy, jejunal conduit, pseudohyperkalaemia; ECG sequence; > **6.0** urgent; binders (SZC ≈1 h, patiromer ≈7 h)*

> **ALGORITHM**
> 1. **Stabilise** if there is any ECG change: calcium gluconate 10%, 10 mL IV.
> 2. **Shift**: insulin 10 U with 50 mL of 50% dextrose; nebulised salbutamol 20 mg.
> 3. **Remove**: haemodialysis if oliguric or ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; a binder.
> 4. Stop culprit drugs.

- **Sodium polystyrene sulfonate (SPS)** is the only emergency-approved binder, but is slow and causes **colonic necrosis** (postoperative, ileus), so it is not first-line.
- **Hypokalaemia** from **vomiting** is caused by kidney loss, not gastric loss. Alkalosis causes bicarbonaturia, which increases distal sodium delivery and so potassium excretion.
- Replace potassium chloride, **oral preferred**, 20–40 mEq repeated.

Calcium

- Saline and loop diuretics raise urine calcium; **thiazides lower it**, because distal tubule calcium handling is not sodium-coupled.
- Corrected calcium: add 0.8 mg/dL for each 1 g/dL albumin below 4.
- **Hypercalcaemia in a stone former**: non-suppressed PTH means primary HPT; if suppressed, check PTH-related peptide (renal, bladder, squamous cancer) and 1,25-vitamin D (sarcoid, lymphoma). Low urine calcium suggests familial hypocalciuric hypercalcaemia.
- Treat if symptomatic or ≥ **14 mg/dL**: **isotonic saline** 200 mL/h to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin (rapid, tachyphylaxis 48 h); IV bisphosphonate; denosumab.

Acid-base

- Anion gap (AG) = Na − (Cl + HCO_3); it is raised if >12. Add 2.5 for each 1 g/dL albumin below 4. If the AG rises less than the bicarbonate falls, a non-AG acidosis coexists.
- **Urine AG** = (Na + K) − Cl. Negative means ammonium excretion, so gut loss; positive means a renal cause (renal tubular acidosis, RTA).
- P2 Renal tubular acidosis — *type 1/2/4 table: defect, K, urine pH, stones; obstruction causes type 4; UAG*

Not in the paper answer:

- In type 4, hyperkalaemia itself lowers ammoniagenesis, which deepens the acidosis.
- Proximal RTA: hypokalaemia worsens with bicarbonate; urine pH is high on carbonic anhydrase inhibitors. **Topiramate and zonisamide** cause calcium phosphate stones.

- Hyperkalaemic distal RTA: **obstruction**, lupus, sickle cell, amiloride, triamterene, trimethoprim. 1 mmol citrate equals 1 mmol bicarbonate.
- MCQ Laxative abuse gives a negative urine AG and urine pH >5.8, which leads to **ammonium acid urate stones**.
- †Ileal or colonic conduits and neobladders absorb ammonium chloride, which causes a hyperchloraemic non-AG acidosis.

8 · Urinary tract obstruction and post-obstructive diuresis

Obstruction is curable, but its relief can cause dangerous fluid loss.

Term	Meaning
Hydronephrosis	Pelvicalyceal dilatation, with or without obstruction
Obstructive uropathy (OU)	Obstruction at any level
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **curable**, so exclude it (†by ultrasound) in any unexplained AKI or CKD. Unilateral OU may not raise SCr *see Paediatric Urology book, Ch 51*.
- P1 Post-obstructive diuresis — *after bilateral obstruction relief; up to **300 mL/h**, ≈2 days; 3 mechanisms (AQP2/Na-K-ATPase loss, volume expansion/ANP, urea); gradual decompression no role*

Not in the paper answer:

- Transporter suppression may last weeks, causing nephrogenic DI with urine 300–400 mOsm/kg and hypernatraemia.

ALGORITHM

1. Relieve the obstruction, then allow free oral fluids; do not give routine IV fluid.
2. If output stays high, or there is hypovolaemia, or the patient is obtunded or nil by mouth, then check a panel every **4–6 h** (Na, K, phosphate, Ca, Mg).
3. Replace ≈**75%** of the output with **0.45% saline** plus electrolytes.

KEY POINT

Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement keeps the patient expanded and sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete unilateral ureteral obstruction	100% at 7 days; 70% at 14 days; 30% at 4 weeks; **none at 6 weeks**

- Injury is **vascular** before tubular; duration, not dilatation, decides it.
- A small stone with low-grade obstruction can be observed safely for **2–4 weeks**. Beyond 4 weeks outcome is unpredictable (recovery up to 5 months).
- Recovery cannot be judged acutely: keep a **low threshold for a stent or nephrostomy**. †Infection with obstruction needs urgent drainage.

9 · Landmark trials

These trials set current practice.

Trial	Design	Result	Take-home
†PRESERVE	Bicarbonate vs saline; N-acetylcysteine (NAC) vs placebo	No difference	Saline; no NAC
†STARRT-AKI	Accelerated vs standard renal replacement therapy	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	SGLT2 inhibitor, CKD with or without diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	Finerenone, CKD with type 2 diabetes	Fewer kidney and cardiovascular events	Add to RAAS blockade
†CHOIR / TREAT	High vs lower haemoglobin target	Harm with high target	Avoid Hb >13 g/dL

91 Renal Transplantation

1 · End-stage disease and why transplant

Transplant beats dialysis, so plan it early.

- KDIGO stage **G5 (eGFR < 15 mL/min/1.73 m²)** is end-stage renal disease (ESRD) *see Ch 90*.
- Dialysis years add cardiovascular and mineral–bone damage, so **transplant is optimal therapy**; **pre-emptive** transplant (before dialysis) is best, commoner with living donors and children.
- Transplant may triple life expectancy and is cheaper after year one; arteriovenous access is still made ahead of need. Death risk is **higher for the first 3–6 months**, then much lower (Wolfe).
- MCQ First successful kidney transplant: identical twins, 1954.

2 · Recipient evaluation

Work-up screens out those surgery or immunosuppression would harm.

> **KEY POINT**
> Two questions: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Evaluate before dialysis; deceased-donor listing is allowed at eGFR ≤ **20**.

Recurrence risk	Diseases	Implication
Very low	Autosomal dominant polycystic kidney disease (ADPKD), dysplasia, Alport, **genetic** focal segmental glomerulosclerosis (FSGS)	None
Higher	**Primary FSGS**, membranoproliferative GN, membranous	Counsel; monitor
Histological often, clinical rarely	IgA nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Disease coded in native-kidney genes cannot recur in a graft carrying donor genes; disease from circulating or immune factors can.

- **Cardiac**: coronary artery disease is the main cause of death. High-risk patients (diabetes, smoking, prior coronary disease, long dialysis) need echo, stress test and selective angiography.
- **Vascular**: heavy iliac calcification may prevent anastomosis (Doppler, non-contrast CT). A non-healing foot ulcer defers listing. ADPKD with family history of intracranial bleed needs an aneurysm screen.
- **Obesity** (relative contraindication) increases wound complications, delayed graft function (DGF) and rejection. Frailty predicts mortality. Non-adherence and substance abuse must be controlled.
- **Infection**: active tuberculosis contraindicates until treated. Screen hepatitis B, C and HIV; cytomegalovirus, herpes simplex and Epstein–Barr serology guides prophylaxis.

- **Vaccinate before**: live vaccines are barred afterwards. Antivirals allow hepatitis C-viraemic kidneys into negative recipients.

Malignancy	Transplant
Prostate Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Without prior treatment
Basal/squamous skin, melanoma in situ, renal cell carcinoma <3 cm, small low-grade thyroid, superficial bladder cancer	No wait after cure
All other	After treatment, individualised wait

- After long-term femoral dialysis access, exclude iliac venous obstruction; a same-side leg fistula causes graft venous hypertension.
- P2 Pre transplant nephrectomy — *indications: ADPKD space/infection/bleeding/pain, chronic infection, VUR with UTI, heavy proteinuria*

3 · Immunological compatibility

Blood group and HLA matching prevent immediate rejection.

- P3 Discuss the role of tissue typing and cross match in live and cadaveric — *ABO rules (A2 donors), HLA class I/II, sensitisation, crossmatch, Luminex virtual crossmatch, paired exchange*

4 · Deceased donation

Donor type and preservation decide early graft function.

- P2 P1 Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval; Kidney retrieval techniques for organ donation — *DBD vs DCD, contraindications, KDPI/EPTS, en bloc <**15 kg**, retrieval steps, UW/HTK, machine perfusion ↓DGF*

Not in the paper answer:

- En bloc kidneys go to an adult with body mass index <30 and low immunological risk; thrombosis is 10–15%, but long-term survival equals living donation.
- Cold storage at 4 °C slows, not stops, metabolism. Machine perfusion acts on ischaemia–reperfusion injury, with no long-term graft benefit.

	Donation after brain death (DBD)	Donation after circulatory death (DCD)
Warm ischaemia	Minimal	Present, so rapid aortic cannulation and cooling are needed
Preservation	Static cold storage	Usually machine perfusion (reduces DGF)

ALGORITHM

1. Midline incision and sternotomy; mobilise bowel off the retroperitoneum.
2. Control the aorta and inferior vena cava (IVC).
3. Find aberrant or accessory arteries **before cannulation**.
4. Perfuse; ice via opened Gerota fascia.
5. Remove kidneys en bloc with **periureteric tissue preserved**.
6. Separate on the back table; store cold.

5 · Living donation

A healthy person takes the risk, so donor safety comes first.

- P2 Pre-operative evaluation of the living (related and unrelated) renal donor — *donor safety, exclusions (bilateral stones), GFR by 2 methods, CT angiography, split function; function 60–80%, ESKD 90 vs 14/10,000*

Not in the paper answer:

- General-population lifetime ESRD is 326/10,000.
- Donors have more gestational hypertension and pre-eclampsia.
- Donor mortality is **3/10,000**. An independent donor advocate is mandatory.
- **Laparoscopic donor nephrectomy** (Ratner 1995) is standard. Accessory arteries may contraindicate it *see Ch 88*.

6 · The operation

Technical errors cause most early graft losses.

- Check ABO before induction. On the back table, ligate vein tributaries and hilar lymphatics (lymphocele). **Never dissect the triangle between lower pole and ureter.** Extend a short right vein with donor IVC.
- **Multiple arteries** (18–43%) mean more DGF but similar survival; use a Carrel patch, conjoining, reimplantation or separate anastomoses.
- **Always keep the lower pole artery**, because it supplies the ureter; upper pole branches ≤2 mm may be ligated.
- The graft goes in the retroperitoneal iliac fossa (right traditional). **Contralateral placement** brings the pelvis and ureter anterior.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near the anterior superior iliac spine (ASIS)	Midline suprapubic, up the rectus edge
Access	Iliac vessels	Better to common iliac, IVC
Hernia	Lower	Higher; rectus denervation

- Nerves at risk: lateral femoral cutaneous (near ASIS), femoral (retractor), genitofemoral.

ALGORITHM

1. **Vein first**: end-to-side to the external iliac vein, 5-0/6-0 polypropylene.
2. Artery end-to-side to the external iliac artery (punch arteriotomy; avoid calcified clamp sites).
3. Reperfuse, then position the kidney without kink or stretch.

- **Hume test**: finger compression of the renal vein causes pulsatile swelling, relieved on release.
- A flaccid graft suggests poor inflow; a tense, engorged graft, poor outflow. Check with intraoperative ultrasound.
- Avoid the internal iliac if the opposite one is occluded (gluteal claudication, erectile dysfunction).

ALGORITHM

1. **Lich–Gregoir** extravesical: fill the bladder; incise the detrusor down to mucosa.
2. Open the mucosa distally and spatulate the ureter.
3. Suture mucosa to mucosa with 5-0/6-0 absorbable, with or without a stent.
4. Close the detrusor loosely as a short tunnel.

- Document stent removal; forgotten stents encrust. Urological complications *see Ch 92*.

7 · Surgical complications

Early vascular problems lose grafts within hours.

- P1 Vascular complications after renal transplantation, including transplant renal artery stenosis — *arterial/venous thrombosis (reversed diastolic flow), TRAS (parvus-tardus, angioplasty first), thrombosis risks, torsion, pseudoaneurysm, biopsy AV fistula*

Not in the paper answer:

- Bleeding (brisk drain, falling haematocrit): transfuse, re-explore.
- P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — *risks incl. mTOR inhibitors; lymph creatinine = serum; treat if symptomatic; laparoscopic fenestration best, aspiration worst*

Transplant nephrectomy

- Indications: **graft intolerance syndrome** (pain, swelling, haematuria, fever; steroids first), infection, tumour; not an asymptomatic failed graft.
- **Early (<6–8 weeks), go extracapsular**; later, go **subcapsular**, except for tumour.

8 · Rejection

Chronic antibody injury now limits graft survival.

- The usual regimen is induction, then tacrolimus and mycophenolate, with or without steroids.
- P1 Renal allograft rejection. — *hyperacute/acute/subclinical/chronic; acute <10%, chronic AMR limits survival; biopsy (Banff TCMR/AMR); risks; de novo DSA, dd-cfDNA NPV >90%; TCMR steroids, then ATG; AMR plasmapheresis, IVIG, rituximab, eculizumab, anti-IL-6*

Not in the paper answer:

- De novo donor-specific antibody (DSA) is linked to class II mismatch and under-immunosuppression.
- A young recipient with an old donor kidney is a rejection risk.
- Hyperacute rejection is †irreversible, so the graft is removed.

9 · Landmark trials

Why transplant is offered despite early risk.

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

★ Putting it together: kidney function and transplant decisions

A · Rising creatinine after urological surgery

Exclude obstruction first; it is quickest to fix.

ALGORITHM

1. **Postrenal first:** check the catheter; scan bladder and kidneys. After cystectomy or bladder repair, a rise with ascites may be a urine leak, so drain creatinine ≫ serum confirms it (see Ch 90).
2. **Prerenal vs intrinsic:** if fractional sodium excretion (FeNa) <1% with bland urine, then volume depletion; if >1% with granular casts, then acute tubular necrosis (ATN). Stop nephrotoxins, renin–angiotensin blockers, metformin.
3. **Dialysis** only for refractory "AEIOU": acidosis, electrolytes (K), intoxication, overload, uraemia.

B · Post-obstructive diuresis

Do not chase this diuresis with fluid.

KEY POINT

After bilateral (or solitary-kidney) relief, monitor output and electrolytes. It is mostly physiological; †it is pathological if it persists after euvolaemia. †Do not replace every millilitre. If unable to drink, replace ≈75% with 0.45% saline (see Ch 90).

C · Transplant ureteric complications

Fluid creatinine tells a leak from a lymphocele.

Problem	Clue	Management
Urine leak (early)	High drain output; fluid creatinine ≫ serum	Foley and nephrostomy ± stent; explore if large, proximal or persistent
Ureteric stricture	Falling function ± mild hydronephrosis	Nephrostomy, then balloon dilation or endoureterotomy; open repair (reimplant, ureteroureterostomy, Boari flap) is most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter or vessels	Treat only if symptomatic (<15%); laparoscopic peritoneal window is best; aspiration recurs
Stent-related urinary infection	Colonised stent	Remove at 14–21 days (earlier risks leak; later risks infection)

Prevent with a stent, sparing the lower pole–ureter triangle, and reimplanting lower polar arteries (see Ch 91, 92/61).

D · Rejection at a glance

Biopsy defines the type, which sets treatment.

Type	Cause	Point
Hyperacute	Preformed donor-specific antibody (DSA)	Crossmatch prevents it; †irreversible, so graft nephrectomy
Acute	T-cell-mediated (TCMR) or antibody-mediated (AMR), Banff	Biopsy. TCMR: IV steroids, then antithymocyte globulin (ATG). AMR: steroids, IV immunoglobulin, plasmapheresis, rituximab
Chronic	Mainly antibody-mediated	Main late loss; subclinical

E · Child going for transplant

Fix the bladder before transplant.

ALGORITHM

1. Most have congenital anomalies of kidney and urinary tract (CAKUT), chiefly valves.
2. Make the bladder safe: age-appropriate capacity at detrusor pressure <40 cm H_2O (antimuscarinics, clean intermittent catheterisation, augmentation if needed).
3. Remove refluxing ureters or kidneys causing infection, polyuria or hypertension.
4. A poor bladder destroys the graft (see Ch 92/61).